

**EASTSOUND SEWER AND WATER DISTRICT
EASTSOUND SEWER AND WATER DISTRICT WASTEWATER TREATMENT PLANT
UPGRADE**

**ADDENDUM NO. 4 (13 pages total)
Issue Date: February 9, 2023**

General: Recitations of each and every section of the bid documents impacted by this addendum are not given. The described changes impact all relevant portions of the bid documents whether specifically cited below or not.

CONTRACT CLARIFICATIONS:

1. Surfacing Demolition Information:

Asphalt saw cut information is provided on Sheet C0.5; however, new surfacing information is provided on Sheet C2.6.

2. Exterior Sealant - Precast Concrete Wet Well AND Vault Joints:

Exterior of all precast concrete joints are to be sealed with CANUSA-CPS WrapidSeal (Manhole Encapsulation System), or approved equal. Install per MFR requirements.

- 3. Q:** The prefabricated metal blower building has an exterior and interior factory finish, correct?
A: Yes. See Section 13 12 20, 2.03 Roof Panels and 2.04 Wall Panels.

- 4. Q:** Sheet E9.1 shows the Motor Control Center as a 7 section MCC. Would a larger MCC be acceptable?
A: Yes. The Blower Building footprint dimensions have been adjusted to 9.25' x 44.00'. MCC is to be increased to eight 20" wide sections (160-inches, total length). In addition, see attached Sheet A1.1

5. Temporary Back Up Generator (Bid Item #18) – General Questions:

Q: Rental of Transfer Switch?

A: The transfer switch is to be the permanent ATS for the project which is not a long lead item. The temporary back-up generator is to temporarily connect to the ATS until the permanent genset arrives.

Q: Expectations for testing temporary back-up generator?

A: Test generator connections, automatic load transfer test.

Q: Sound expectations?

A: Quiet operation, 60 dBA or lower at full load (per portable generator specs).

Note: An example cut sheet for a rental generator is attached. See attached **Sunbelt Rentals 120 kW Diesel Generator Exhibit**.

SPECIFICATIONS:

SECTION 09 90 00 MISC PLANT PAINTING, 2.01, I.:

Add the following Service Condition for painting requirements for gypsum wall board that will be provided in the new Blower Building:

8. SERVICE CONDITION J
 - a. Generic Type: Egg-shell alkyd enamel.
 - b. Application: Gypsum board
 - c. Primer: One coat Sherwin-Williams wall primer DF=2.0
 - d. Finish: Two coats Sherwin-Williams Pro-Mar alkyd egg-shell, DF=4.0

SECTION 22 13 36 PACKAGED SLUDGE PUMPS, 2.01 A.

Please adjust language from "Each pump is to include a VFD" to **"Pumps will be powered from VFDs in the Motor Control Center. The VFDs will be controlled by the PLC in the Main Control Panel. Local speed control dial at pump is not required. See Electrical Sheets."**

SECTION 26 80 00 2.04 Instrumentation D. Electromagnetic Flow Meters

Please Note: the specified flow meter (Endress-Hauser, Proline Promag L400 series) has been replaced with model **Proline Promag W400**. W400 is the acceptable alternative.

SECTION 33 41 00 STORM UTILITY DRAINAGE, 2.02: Add the following:

F. Trench Drains

1. Trench drains shall be Zurn Z886-HD or engineer approved equivalent. The channels shall be manufactured of HDPE with a 0.75% built-in slope. The channels shall have an inside width of four inches and the channel inverts shall be radiused. The channels shall be a minimum of 80 inches in length (in order to minimize the amount of joints) with mechanically interlocking ends. The channels will be manufactured with vertical ribs twenty inches on center. Black Dura-coated heavy duty steel frames with concrete embedment studs will be attached to the channels prior to installation. Integral galvanized steel edge rails will not be allowed. Exposed polymer concrete edges will not be allowed.
2. The trench drain grates will be transversely slotted ductile iron with a width of 5.375 inches, and an open area of nominally 18.5 square inches per linear foot of grating. The grates will be locked to the frame with bolts after the final inspection of the system. The trench drain assembly will be rated for Load Class C.

SECTION 43 22 10 2W PLANT WATER SYSTEM EQUIPMENT, 2.01.A

Please adjust language from: "Pumps are to include variable Frequency Drives (VFDs)" to: **"Pumps will be powered from VFDs in the Motor Control Center. The VFDs will be controlled by the PLC in the Main Control Panel. See Electrical Sheets."**

DRAWINGS:

SHEET A1.1, A1.2, C6.4: See edits in red on attached drawing.

Clarification: Blower Building dimensions are to be adjusted as shown. New layout provides approx. 9" clearance between existing Train 2 building and new blower building; however new concrete slab is to extend to the existing Train 2 slab. Maintain 10-ft setback from both south and east property lines. See location adjustments for the MCC, MCP, Bladder Tanks, and Exhaust Fans.

Regarding the metal building configuration, please note the following: The high sidewall will have a semi-flush girt condition with a 5" offset from the outside of the column flange to the outside of Zee girts exterior flange (outside of steel line/foundation). All other sides of the structure will remain in bypass Zee girt condition. Contractor is responsible for means and methods, as needed, to create a weather tight enclosure for north wall. North wall is to include exterior metal wall panels. Wall panels may need to be constructed on the ground and lifted into place.

SHEET C1.3: See edits in red on attached drawing.

Clarification: Additional piping labels have been added. In addition, the 2W (Air Gap) Water Hot Box orientation has been adjusted slightly to allow for vehicle traffic.

SHEET C1.6A: See edits in red on attached drawing.

Clarification: Sludge piping is to be routed below the existing floor in areas clouded in red on attached Sheet C1.6A. Provide concrete coring and/or concrete cutting as needed. All cuts are to be "clean edge" style completed with coring or saw cutting equipment. All concrete floor repair work is to match design requirements (concrete and reinforcement) shown for Blower Building concrete slab. See Detail 8 on Sheet S4.1. New rebar is to be doweled and epoxied into existing concrete with minimum 4" embedment depth into existing concrete.

SHEET S2.3: See clouded edits on attached drawing.

SHEET S4.1: See clouded edits on attached drawing.

ATTACHMENTS INCLUDED ON FOLLOWING PAGES:

- Sheet A1.1 (1 page)
- Sheet A1.2 (1 page)
- Sheet C6.4 (1 page)
- Sheet C1.3 (1 page)
- Sheet C1.6A (1 page)
- Sheet S2.3 (1 page)
- Sheet S4.1 (1 page)
- Sunbelt Rentals 120 kW Diesel Generator Exhibit (3 pages)

1. PROCESS AIR PIPING AND 2W PIPING NOT SHOWN FOR CLARITY, SEE CIVIL SHEETS C1.1, C6.4, AND C6.5 FOR PIPING INFORMATION.

ABBREVIATIONS R.B. = RUBBER BASE EXP STR = EXPOSED STRUCTURE GWB = GYPSUM WALL BOARD AC = ACOUSTICAL WALL BOARD FCS = FIBER CEMENT SIDING TILE = SUSPENDED CEILING TILE PLY = 1/2" PLYWOOD ROOM NAME	ROOM NO.	FLOOR		BASE	WALL		CEILING		
		MAT'L	FINISH	MAT'L	MAT'L	FINISH	MAT'L	FINISH	HEIGHT
BLOWER ROOM	101	CONCRETE	SEALER	6" R.B.	GWB	PAINT	EXP STR	PAINT	VARIABLE

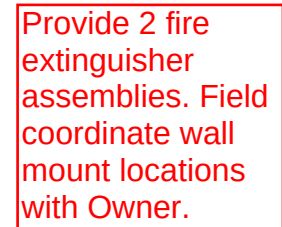
DOOR NO.	DOOR SIZE WxHxT	TYPE	MATERIAL	FINISH	HARDWARE GROUP #		
101A	10'-0"x8'-0"x1 3/4"	B	INS HM	PT	-	FLUSH PANEL	
101B	3'-0"x6'-8"x1 3/4"	A	INS HM	PT	-	TYPE A	TYPE B
101C	5'-0"x8'-0"x1 3/4"	A	INS HM	PT	-	FLUSH PANEL	FLUSH PANEL

FLUSH PANEL

TYPE A

FLUSH PANEL

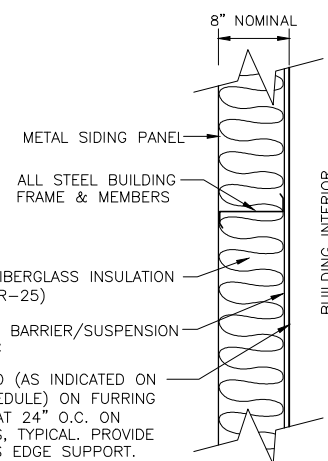
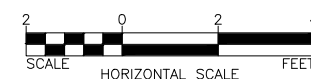
TYPE B



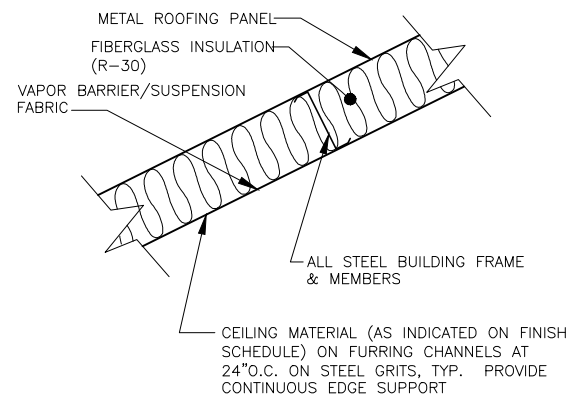
Move 2W bladder tank assembly to new location. See Sheet C6.4.

Field measure existing Train 2 building FF elevation and adjust, as needed, to match. In addition, extend new concrete slab (for blower building) 9" north, abutting the existing Train 2 building slab.

Move MCP Panel to this wall.
Adjust routing of power and
control raceways as required.



1 TYPICAL EXTERIOR WALL
NOT TO SCALE



2 TYPICAL ROOF SECTION
NOT TO SCALE

— CONTRACTOR TO COORDINATE AND ENSURE ALL ELECTRICAL PANELS ARE CONFIGURED WITH 3.5 FOOT CLEARANCE, AS NEEDED, PER ELECTRICAL REQUIREMENTS.

11-02-2022

DESIGNED BY	JGC
DRAWN BY	SNR
CHECKED BY	AWL

EASTSOUND SEWER AND WATER DISTRICT
WASHINGTON COUNTY
WASTEWATER TREATMENT PLANT UPGRADE – PHASE I

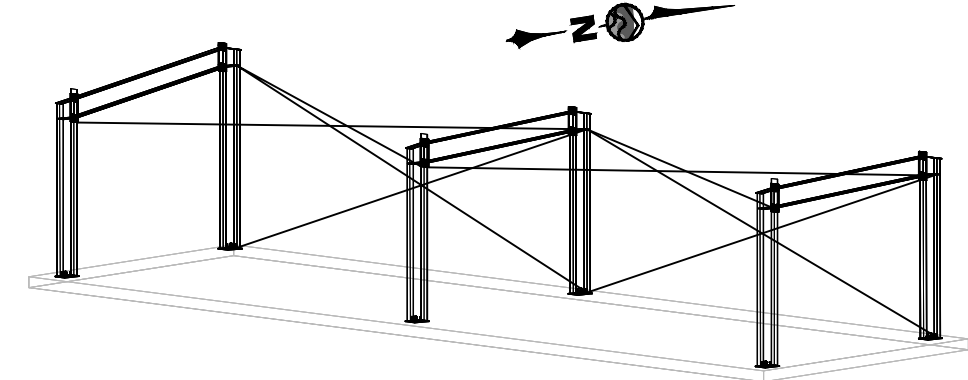
BLOWER BUILDING PLAN

SHEET A1.1	DATE	NOV 2, 2022
	SCALE	AS SHOWN
	JOB NUMBER	2020-070

811 Call 811
two business days
before you dig

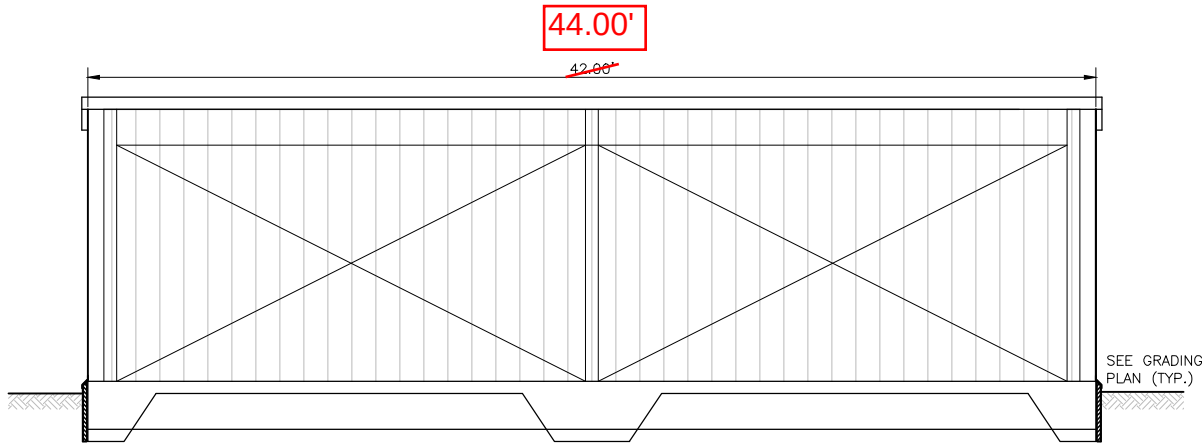
BID SET

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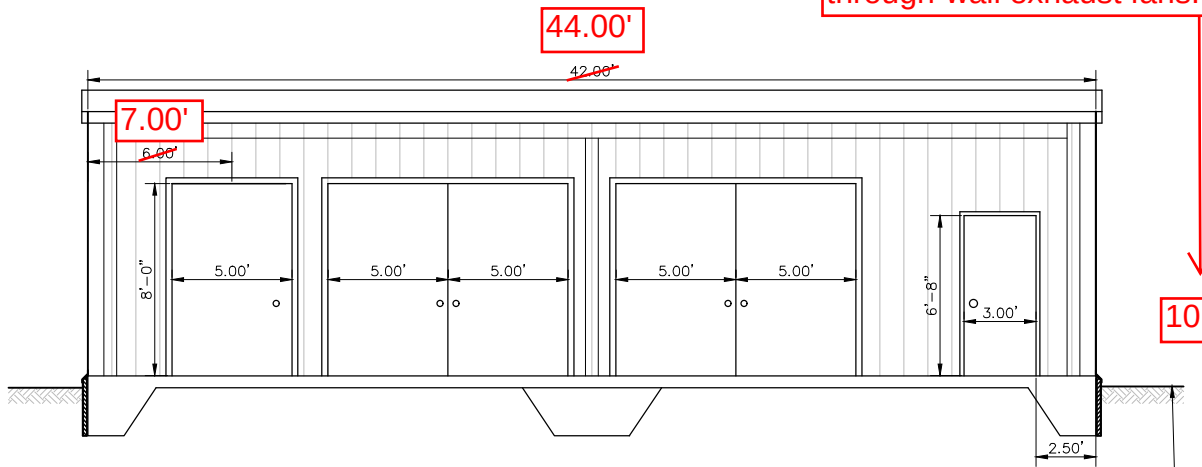
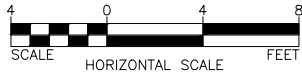


CONCEPTUAL FRAMING
NOT TO SCALE

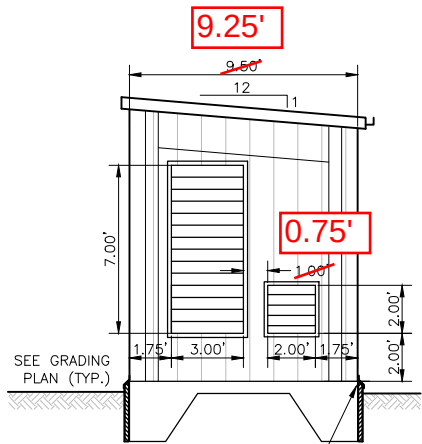
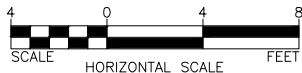
NOTE: CONTRACTOR SHALL PROVIDE
PRE-ENGINEERED METAL BUILDING WITH
SIMILAR COLUMN AND BRACING LAYOUT.



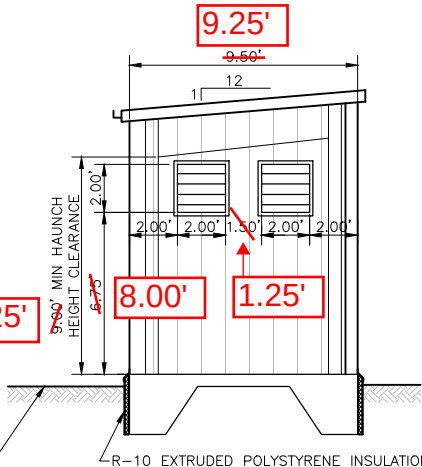
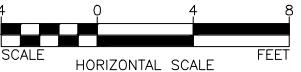
NORTH ELEVATION



SOUTH ELEVATION



WEST ELEVATION



EAST ELEVATION



Increase building height as
shown. Ensure adequate
clearance for both
through-wall exhaust fans.

FLASHING TO COVER CHAMFERED
EDGE OF INSULATION BOARD.
MATCH METAL BUILDING TRIM
COLOR. LAP UNDER SIDING AND
OVER INSULATION BOARD A
MINIMUM OF 3-INCHES (TYP.).

SEE GRADING
PLAN (TYP)

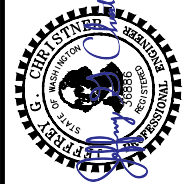
R-10 EXTRUDED POLYSTYRENE INSULATION
BOARD AT FOOTING PERIMETER, FROM TOP
OF CONCRETE TO BOTTOM OF FOOTING.
CHAMFER TOP 45°. COVER EXPOSED
SURFACES WITH GREY DRYVIT EXTERIOR
INSULATION AND FINISH SYSTEM OR EQUAL.



BID SET

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11-02-2022



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DRAWN BY	SNR
CHECKED BY	AWL

EASTSOUND SEWER AND WATER DISTRICT

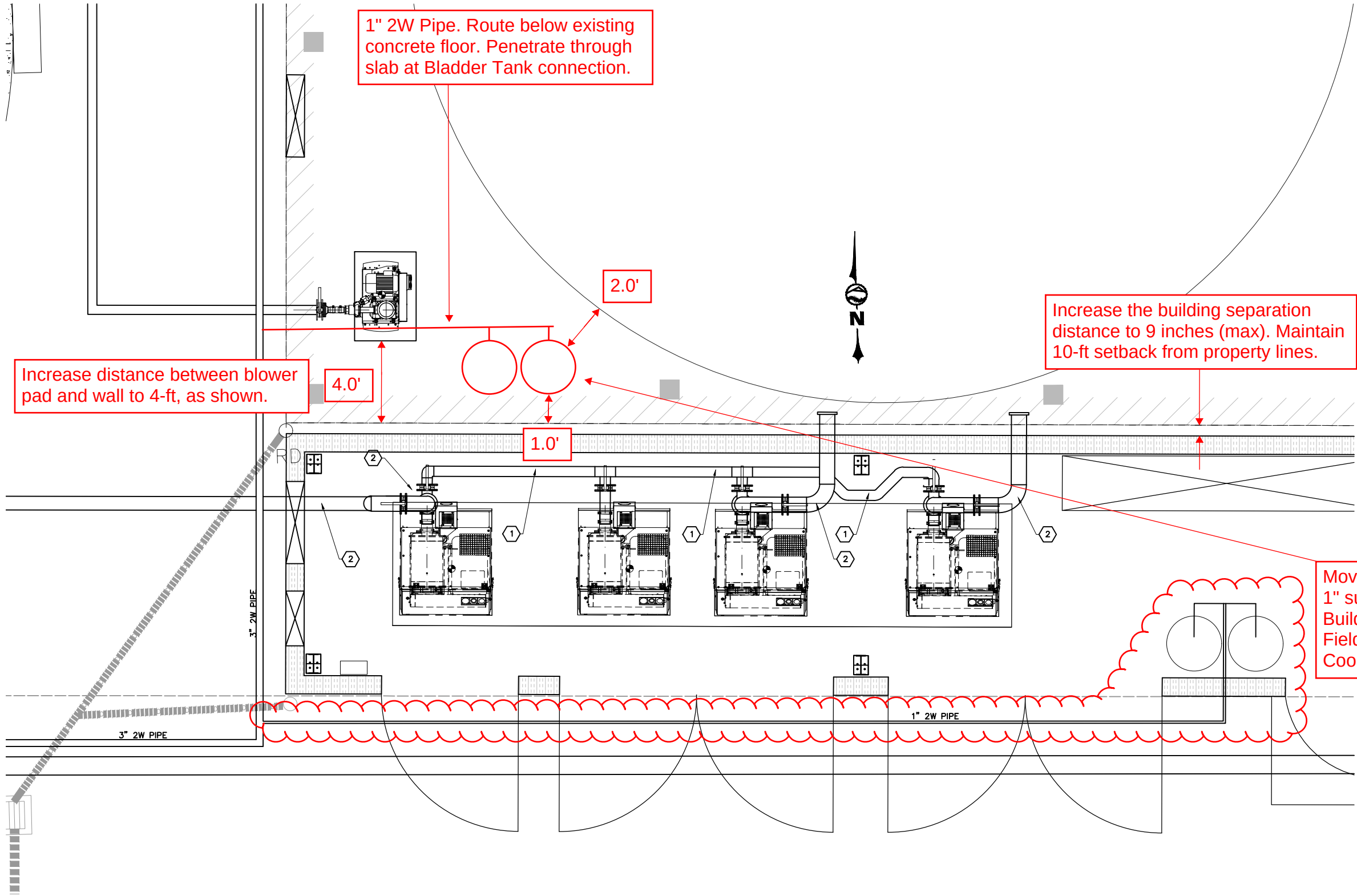
WASHINGTON
WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

BLOWER BUILDING EXTERIOR ELEVATIONS 1

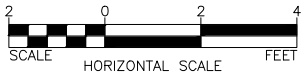
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JOB NUMBER	2020-070

SHEET	A1.2
PAGE	54 OF 96

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W:\2020-070 EASTSOUND WWP DESIGN\DWG\2020-070 C6.4 BLOWERS.DWG - 11/2/2022 3:26 PM - Jeff Smith



BIOLOGICAL TREATMENT BLOWERS



BID SET

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DRAWN BY	JCS
CHECKED BY	AWL

EASTSOUND SEWER AND WATER DISTRICT

WASHINGTON
WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

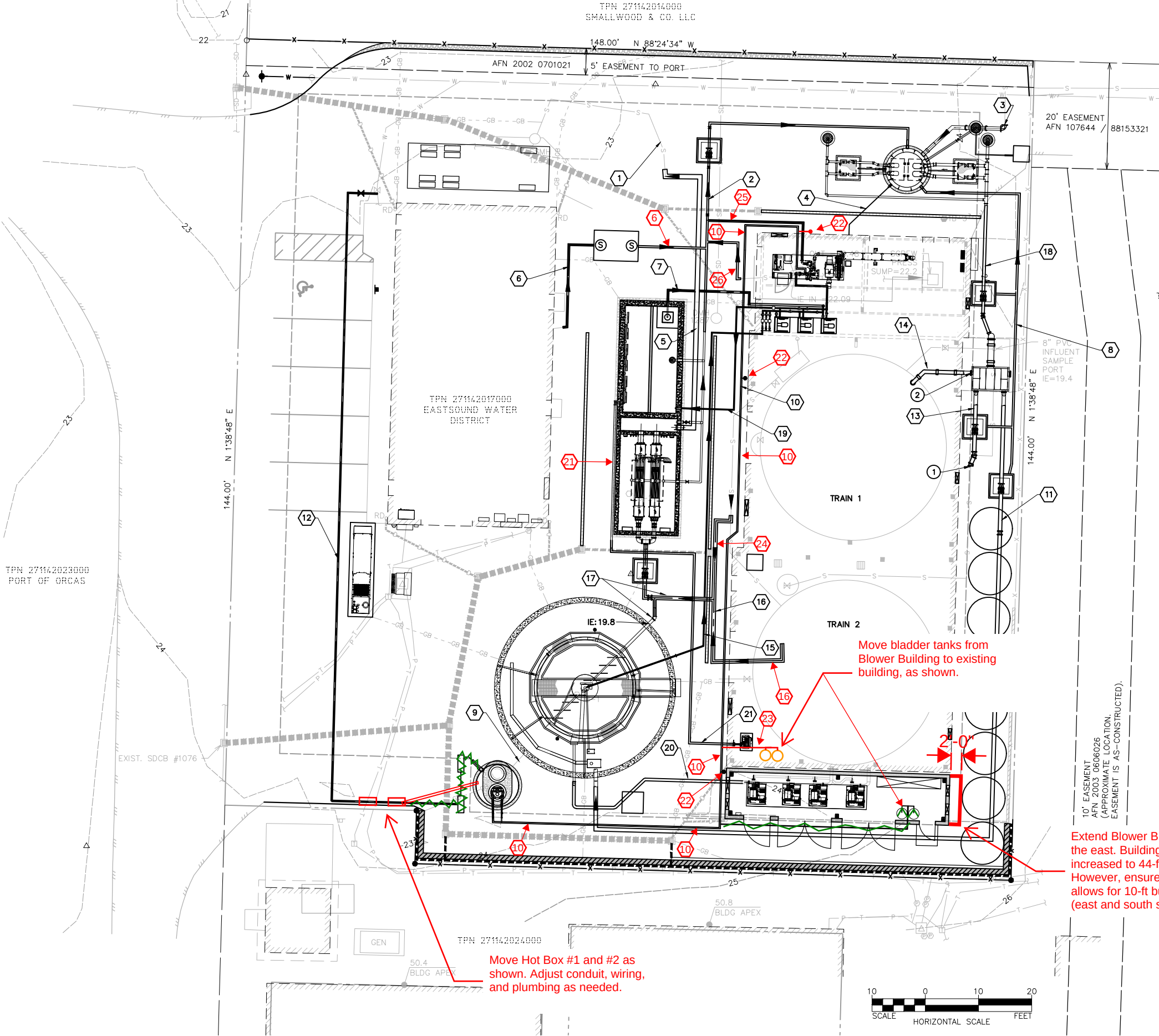
BIOLOGICAL TREATMENT BLOWERS

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DATE	NOV 2, 2022
SCALE	AS SHOWN
JOB NUMBER	2020-070
PAGE	41 OF 96

11-02-2022

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TPN 271142023000
PORT OF ORCAS



PIPE IDENTIFICATION LEGEND	
1	8" EXISTING EFFLUENT (DI GRAV.)
2	4" UV DISINFECTION DRAIN AND DIGESTER SUPERNATANT (DI GRAV.) SLOPE = 2% MIN.
3	8" INFLUENT (DI GRAV.) SLOPE = 0.4% MIN.
4	3" INFLUENT PUMP STATION AIR VENT (SS) OUTLET 8' ABOVE FINISH GRADE WITH GOOSENECK POINTING DOWN. ATTACH PIPE SECURELY TO EXTERIOR BUILDING WALL SLOPE = POSITIVE DRAINAGE THROUGHOUT
5	10" UV DISINFECTION EFFLUENT (DI GRAV.) SLOPE = 0.28% MIN.
6	4" BUILDING SEPTIC (SCH. 80 PVC GRAV.) SLOPE = 2% MIN.
7	2" DIGESTED SLUDGE (SCH. 80 PVC FM) SLOPE = POSITIVE THROUGHOUT
8	4" VAULT DRAIN (SCH. 80 PVC GRAV.) SLOPE = 2.0%
9	1" 2W WATER SYSTEM AIR VENT (SS) OUTLET 3' ABOVE GRADE WITH GOOSENECK POINTING DOWN SLOPE = POSITIVE DRAINAGE THROUGHOUT
10	2" 2W WATER LINE (SCH. 80 PVC FM) HDPE
11	8" TRAIN 3 INFLUENT (DI GRAV.) SLOPE = 0.4% MIN.
12	2" WATER PIPE (SCH. 80 PVC FM)
13	8" TRAIN 2 INFLUENT (DI GRAV.) SLOPE = 0.4% MIN.
14	8" TRAIN 1 INFLUENT (DI GRAV.) SLOPE = 0.4% MIN.
15	2" TRAIN 3 WAS (SCH. 80 PVC GRAV.) SLOPE = POSITIVE DRAINAGE THROUGHOUT
16	8" TRAIN 2 EFFLUENT (DI GRAV.) SLOPE = 0.4% MIN.
17	8" TRAIN 3 EFFLUENT (DI GRAV.) SLOPE = 0.4% MIN.
18	6" INFLUENT TO FLOW SPLITTER (DI FM) SLOPE = POSITIVE THROUGHOUT
19	2" DIGESTER WAS INFLUENT (SCH. 80 PVC GRAV.)
20	6" CELL 3 AIR (SS)
21	4" DIGESTER AIR (SS)

- 22 1" YARD HYDRANTS (3 LOCATIONS)
- 23 1" 2W WATER LINE HDPE

KEYED NOTES

- 1 = CUT EXISTING 8" PIPE AND CONNECT NEW 8" INFLUENT PIPE TO TRAIN 2
- 2 = EXTEND NEW 8" PIPE TO TRAIN 1. ABANDON AND PLUG EXISTING TRAIN 1 INFLUENT PIPE
- 24 8" TRAIN 1 EFFLUENT (DI GRAV.) SLOPE = 0.4% MIN
- 25 4" PRESSATE DRAIN (DI GRAV.) SLOPE = 2% MIN
- 26 4" BUILDING FLOOR DRAIN (DI GRAV.) SLOPE = 2% MIN

SHEET NOTES

- 1) SEE SHEET C0.1 FOR EXISTING CONDITIONS.
- 2) SEE SHEETS C2.1-C2.4 FOR STORM GRADING, DRAINAGE, AND PIPING.

Extend Blower Building 2-ft to the east. Building length to be increased to 44-ft (not 42-ft). However, ensure layout still allows for 10-ft building setback (east and south sides).

Move Hot Box #1 and #2 as shown. Adjust conduit, wiring, and plumbing as needed.



BID SET

NO.	REVISIONS	BY	DATE
1	ADD INF. SAMPLE SUMP	JGC	5/10/22



DESIGNED BY	JGC	DRAWN BY	SNR	CHECKED BY	AWL
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EASTSOUND SEWER AND WATER DISTRICT
WASHINGTON
WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

SHEET	DATE	SCALE	JOB NUMBER
C1.3	NOV 2, 2022	AS SHOWN	2020-070
PAGE			17 OF 96

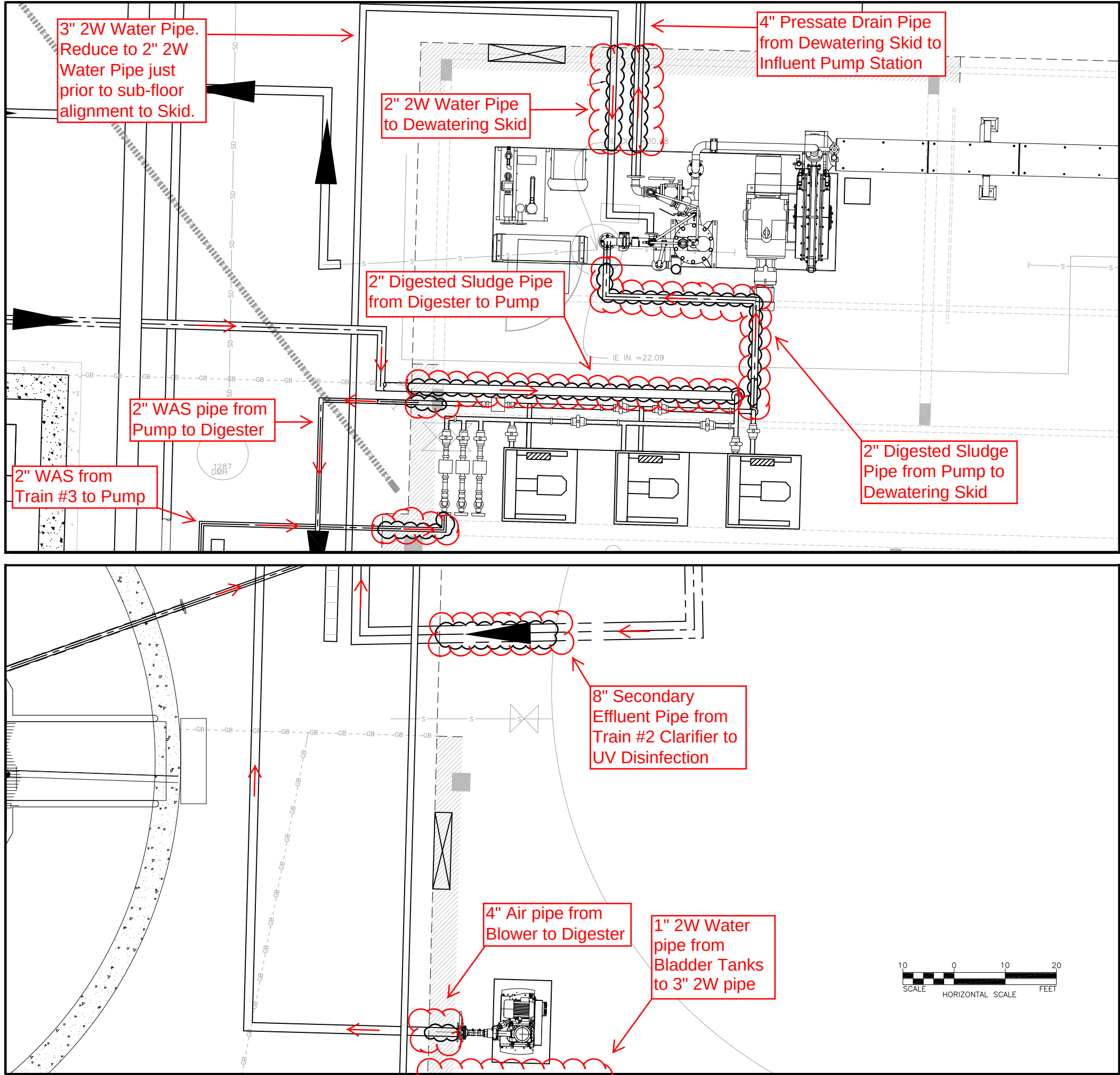
SITE PIPING PLAN WITH PIPING ID

WILSONENGINEERING.COM

11-02-2022

PLOT SETTINGS: Adobe PDF, Ledger, Portrait, 1:2, WE APWA_SCREENED.ctb

W:\2020-2070 EASTSOUND WWP DESIGN\DWG\2020-070 C1.6A SLUDGE PIPING LAYOUT IN BUILDING.DWG - 1/26/2023 5:46 PM - Sam Rodriguez



NOTE:
ALL CLOUED PIPES ARE TO BE
INSTALLED BELOW EXISTING BUILDING
CONCRETE SLABS. CONTRACTOR TO
USE CONCRETE CORING AND/OR
CONCRETE SAW EQUIPMENT TO
PROVIDE ACCESS FOR SUB FLOOR
PIPING WORK. SEE ADDENDUM #3.

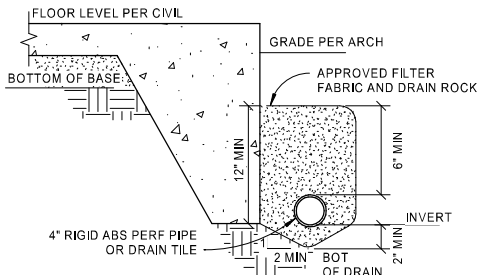


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SHEET C1.6A	DATE NOV 2, 2022	EASTSOUND SEWER AND WATER DISTRICT		DESIGNED BY JCC
	SCALE AS SHOWN	SAN JUAN COUNTY WASHINGTON		
PAGE XX	JOB NUMBER 2020-070	WASTEWATER TREATMENT PLANT UPGRADE - PHASE I		CHECKED BY AWL
OF 96		BURIED PIPING IN BUILDING		

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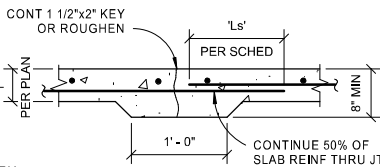
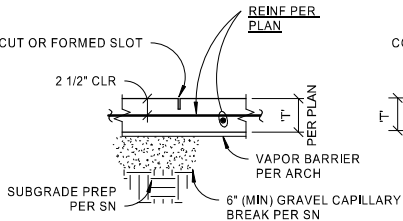
PERIMETER FOUNDATION DRAINS
- INVERT MUST BE BELOW FLOOR LEVEL.
- BOTTOM OF DRAIN MUST BE BELOW BOT OF BASE.

- NOTE**
1. DRAIN PIPES MUST DISCHARGE BY GRAVITY OR MECHANICAL MEANS TO AN APPROVED DRAINAGE SYSTEM THAT COMPLIES WITH THE INTERNATIONAL PLUMBING CODE.
 2. MINIMUM SLOPE FOR DRAIN PIPE RUNS IS 0.25%.
 3. DO NOT CONNECT FOUNDATION DRAIN SYSTEMS TO RAIN LEADERS OR DOWN SPOUTS.
 4. PROVIDE APPROVED CLEANOUTS FOR ALL DRAIN PIPE AT ENDS, CORNERS, AND NO MORE THAN 120'-0" ON CENTER.
 5. SEE CIVIL DRAWINGS FOR GRADES, DISCHARGE POINTS, AND OTHER REQMTS. ALSO SEE GEOTECH REPORT FOR ADDITIONAL REQMTS.
 6. SUBMIT A FOUNDATION DRAINAGE PLAN TO CIVIL FOR REVIEW.
 7. DRAIN ROCK/BASE MATERIAL SHALL CONSIST OF GRAVEL OR STONE WITH LESS THAN 10% OF MATERIAL PASSING THROUGH A NUMBER 4 SIEVE.

7 TYP FOUNDATION DRAINS AT BLOWER BUILDING SLAB EDGE
1" = 1'-0"

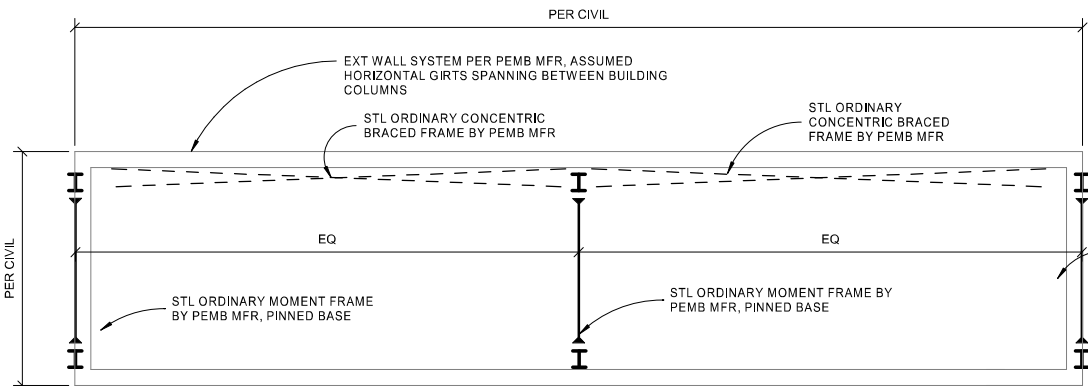
TYP CJ: T/4 DEPTH SAW CUT OR FORMED SLOT

TYP CONTROL JT AND SUBGRADE



- NOTE**
1. PLACE CONTROL JOINTS AND CONST JOINTS AT NO MORE THAN 12'-0" O/C, EW. ALIGN JOINTS WITH COLS AND SLAB CORNERS WHERE POSSIBLE.
 2. CUT CJs AFTER CONC HAS SUFFICIENTLY HARDENED TO PREVENT SPALLING OR TEARING BUT NO MORE THAN 12 HOURS AFTER POUR.
 3. SUBMIT A SLAB PLAN SHOWING CONTROL & CONSTRUCTION JOINT LAYOUT AND POUR SEQUENCE FOR APPROVAL.
 4. THE SUBGRADE SHOWN ABOVE SHALL BE THE MINIMUM REQUIRED AT ANY LOCATION, AND SHALL BE INCREASED OR MODIFIED AS RECOMMENDED IN THE GEOTECH REPORT.
 5. SEE SPEC AND ARCH/MECH/CIVIL DRAWINGS FOR SLAB EXTENTS, SLOPES, STEPS, DRAINS, ELEVATIONS, CHAMFERS, FINISHES AND OTHER RELATED INFORMATION. MAINTAIN MIN THICKNESS PER PLAN THROUGHOUT.
 6. WHERE SINGLE PENETRATION (OR COMBINED GROUPING OF MULTIPLE PENETRATIONS, SUCH AS AT GROUPS OF PIPES OR CONDUIT) LARGER THAN 0'-10" ANY DIRECTION, PROVIDE ADD'L #3 EA SIDE EXTENDING 18" MIN BEYOND THE PENETRATION EXTENTS.
 7. AT INT CORNERS, PROVIDE (2) ADD'L #3x36" @ 2" O/C, PLACED AT 45-DEG AND 2" CLR TO THE INT CORNER OF SOG.
 8. SIM AT EXT SOG'S/SIDEWALKS AROUND BLDG, EXCEPT OMIT VAPOR BARRIER, REDUCE CJ SPACING TO 10'-0" O/C MAX, PROVIDE AIR ENTRAINMENT PER SN.

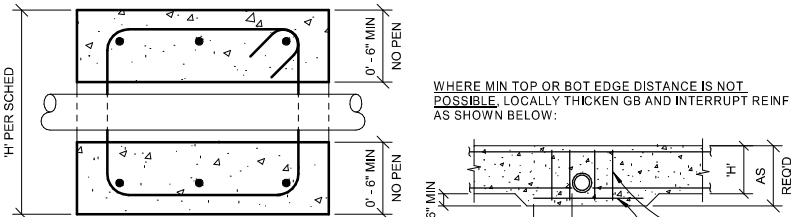
5 TYP SLAB-ON-GRADE DETAILS
1" = 1'-0"



NOTE: THE LAYOUT SHOWN HERE WAS USED AS THE BASIS OF DESIGN FOR THE PEMB FOUNDATIONS, ORIENT LATERAL SYSTEM AS SHOWN.

2 BLOWER BLDG - PEMB LATERAL SYSTEM LAYOUT
1/4" = 1'-0"

- NOTE**
1. PROVIDE SLEEVES W/ ID 2.5" LARGER THAN PIPE OD. CENTER PIPE AND WRAP W/ COMPRESSIBLE MATERIAL.
 2. ALL PROPOSED GB PENETRATIONS SHALL BE COORDINATED BY THE GC. PROVIDE 6" MIN CONC BETWEEN ADJACENT PENETRATIONS. CLEARLY INDICATE ALL PENETRATIONS ON THE SUBMITTED REINF SHOP DRAWINGS: THE ENGINEER SHALL REVIEW THE SHOP DRAWING FOR ADEQUACY AND ADD REINF AS REQUIRED. FOR BIDDING PURPOSES ONLY, ASSUME THAT ALL CONT LONG REINF IS LAP SPLICED WITH EQUIVALENT BAR AREA.
 3. SECTIONS SHOWS ACCEPTABLE PENETRATION LOCATIONS THROUGH AND BENEATH GB'S.
 4. DO-NOT INTERRUPT REINF IN ANY CASE. MAINTAIN COVER PER STRUCTURAL NOTES AT ALL REINFORCEMENT.



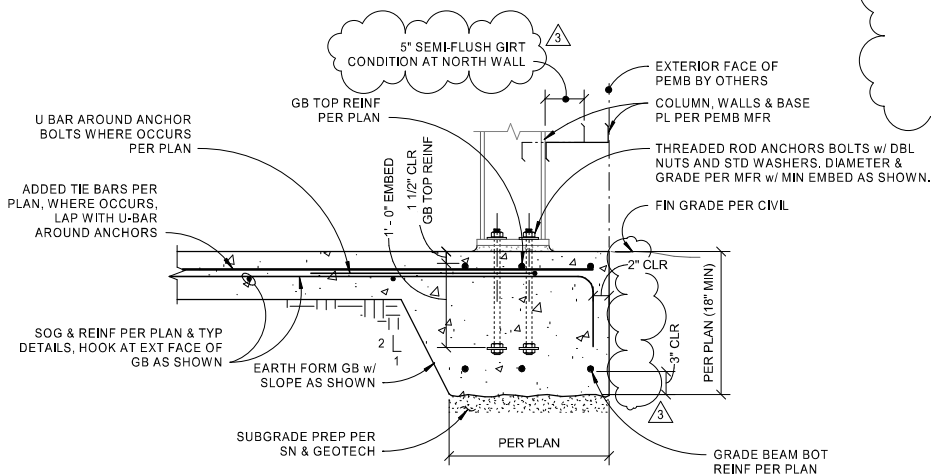
TYP SECTION

WHERE MIN TOP OR BOT EDGE DISTANCE IS NOT POSSIBLE, LOCALLY THICKEN GB AND INTERRUPT REINF AS SHOWN BELOW:

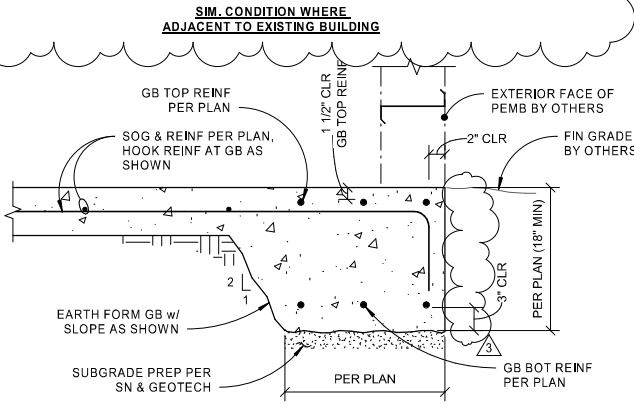
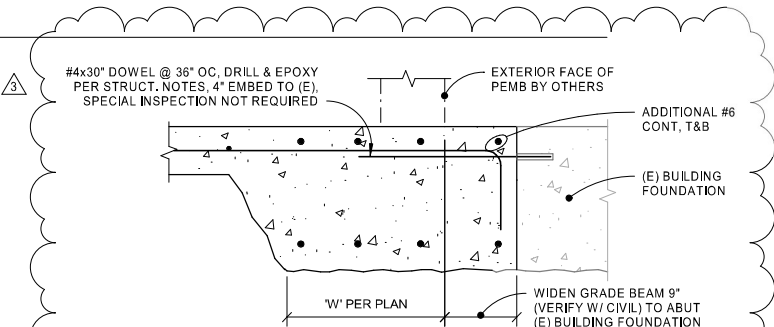
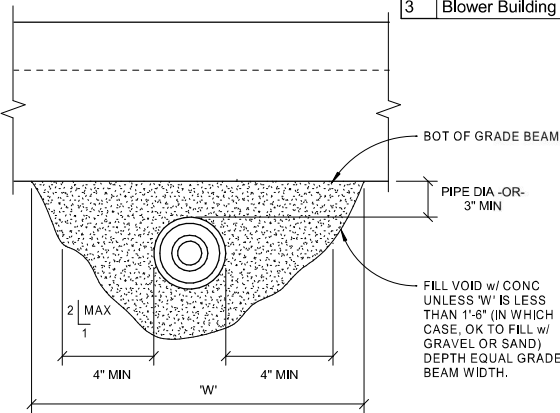


WHERE TYP CONT LONG REINF IS INTERRUPTED, ADD EXTRA LONG BARS W/ SAME REINF AREA AS INTERRUPTED BARS. 'LD' SHALL BE PER SCHED ACCORDING TO THE SIZE OF ADDED BARS.

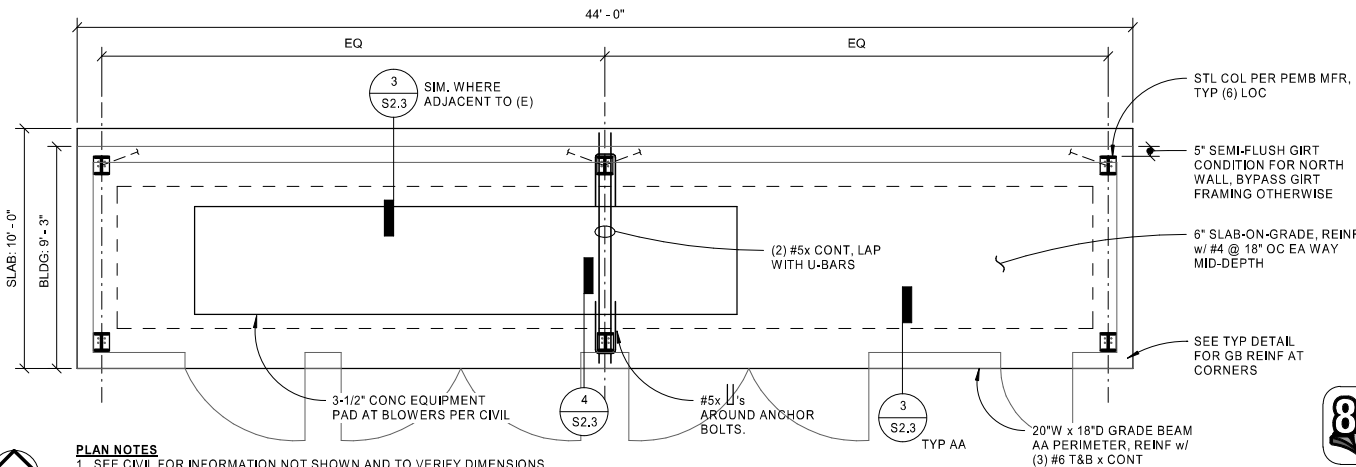
6 TYP GB PENETRATIONS
1 1/2" = 1'-0"



4 TYP GRADE BEAM SECTION AT COLUMNS
1" = 1'-0"



3 TYP GRADE BEAM SECTION
1" = 1'-0"



- PLAN NOTES**
1. SEE CIVIL FOR INFORMATION NOT SHOWN AND TO VERIFY DIMENSIONS.
 2. PRE ENGINEERED METAL BUILDING (PEMB) IS DESIGNED BY OTHERS. SEE DEFERRED SUBMITAL SECTION OF STRUCTURAL NOTES AND SPECIFICATION. FOUNDATION DESIGN IS SUBJECT TO CHANGE AFTER PEMB DESIGN IS FINALIZED.

1 BLOWER BLDG FOUNDATION PLAN
1/4" = 1'-0"



NO.	REVISIONS	BY	DATE
1	UV Disinfection	TH	5/4/22
2	Bid Set	TH	9/6/22
3	Blower Building	TH	2/8/23

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Verify Scale:
Enclosing box measures 1/2 inch x 2 inch wide
when drawings are printed at full scale.

DESIGNED BY
Blomson

DRAWN BY
Ckinney

CHECKED BY
Blomson

EASTSOUND SEWER AND WATER DISTRICT

WASHINGTON

WASTEWATER TREATMENT PLANT UPGRADE - PHASE 1

BLOWER BUILDING FOUNDATION PLAN & DETAILS

DATE
10-15-2021

SCALE
As indicated
(on 22'x34')

JOB NUMBER
Wilson: 2020-070
KW: 211131

SHEET
S2.3

PAGE
59

OF
96

Call 811
two business days
before you dig

BID SET

NO.	REVISIONS	BY	DATE
1	UV Disinfection	TH	5/4/22
2	Bid Set	TH	9/6/22
3	Blower Building	TH	2/8/23



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DRAWN BY
Ckinney

CHECKED BY
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EASTSOUND SEWER AND WATER DISTRICT

SAN JUAN COUNTY WASHINGTON

WASTEWATER TREATMENT PLANT UPGRADE - PHASE 1

TYP STRUCTURAL DETAILS

SHEET
S4.1

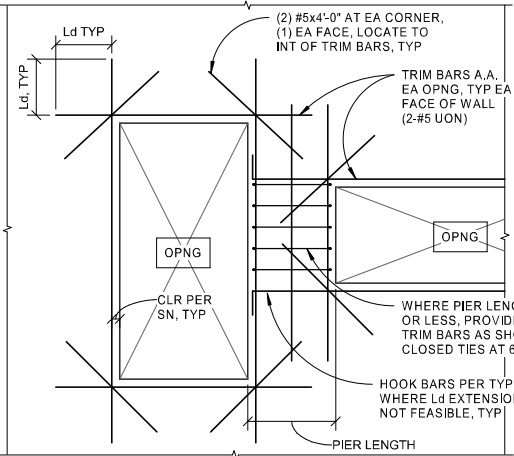
SCALE
As indicated
(on 22"x34")

PAGE
61

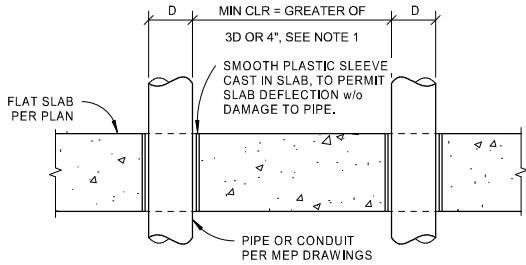
OF
96



BID SET



- NOTES**
1. DETAIL APPLIES TO ALL OPENINGS WITH EITHER HEIGHT OR WIDTH GREATER THAN 2'-0".
 2. Ld IS DEVELOPMENT LENGTH PER TYPICAL DETAILS.
 3. REINF SHOWN ABOVE IS TYP MINIMUM. OMIT ADDED REINFORCEMENT ABOVE WHEN SPECIAL REINF INDICATED ON PLANS OR DETAILS EXCEEDS THIS REINF.
 4. CONTRACTOR SHALL VERIFY ALL OPENINGS NOT SHOWN IN THE STRUCTURAL DRAWINGS WITH THE STRUCTURAL ENGINEER PRIOR TO SUBMITTAL DEVELOPMENT AND INSTALLATION.



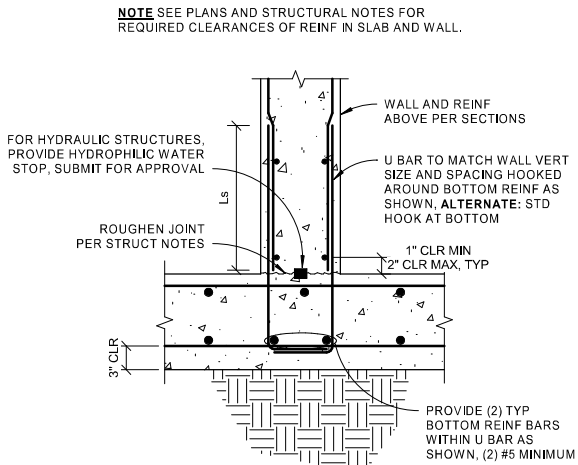
- NOTES**
1. IF MINIMUM CLR SPACING CANNOT BE ACHIEVED, REINF AROUND COMBINED EXTENTS PER TYP LARGE OPENING DETAIL.
 2. DETAIL APPLIES TO D= 9" OR LESS.

6 TYP SMALL PENETRATIONS IN CONCRETE
3/4" = 1'-0"

BAR SIZE	DEVELOPMENT LENGTHS (Ld)		LAP SPlice LENGTHS (Ls)	
	TOP	OTHER	TOP	OTHER
#3	16 in	12 in	18 in	16 in
#4	20 in	15 in	24 in	20 in
#5	28 in	22 in	36 in	28 in
#6	38 in	30 in	50 in	38 in
#7	60 in	46 in	78 in	60 in
#8	74 in	57 in	96 in	74 in
#9	90 in	69 in	116 in	90 in
#10	108 in	83 in	140 in	108 in
#11	126 in	97 in	164 in	126 in

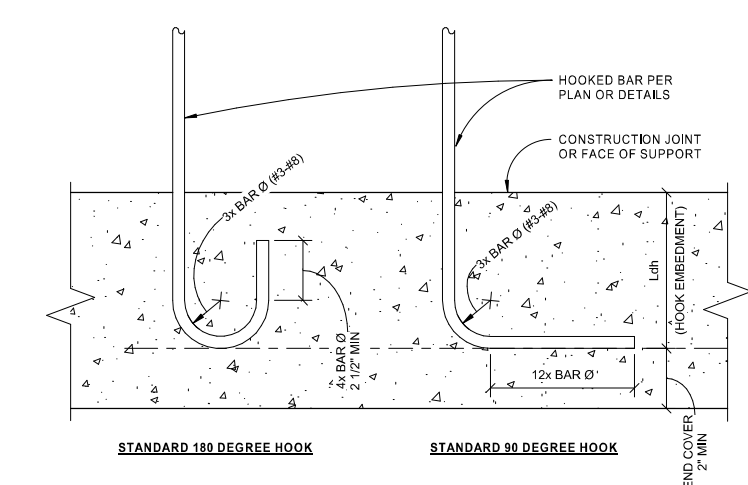
- NOTES**
1. TABULATED VALUES ARE APPLICABLE FOR GRADE 60 UNCOATED REINF BARS.
 2. BAR C/C SPACING MUST BE GREATER THAN TWICE THE BAR DIAMETER AND COVER GREATER THAN ONE BAR DIAMETER.
 3. "TOP" BARS ARE DEFINED AS HORIZONTAL BARS WITH MORE THAN 12 INCHES OF FRESH CONCRETE CAST BELOW THE BARS, INCLUDING WALL HORIZONTALS.
 4. AT LOCATIONS WHERE Ld IS CALLED OUT BUT CANNOT BE ACHIEVED, EXTEND BAR AS FAR AS POSSIBLE AND PROVIDE STANDARD HOOK PER TYP DETAIL.

3 TYP CONCRETE REINFORCING DEVELOPMENT / SPlice LENGTHS
3/4" = 1'-0"



1 TYP CONCRETE WALL REINFORCING TO MAT SLAB
1" = 1'-0"

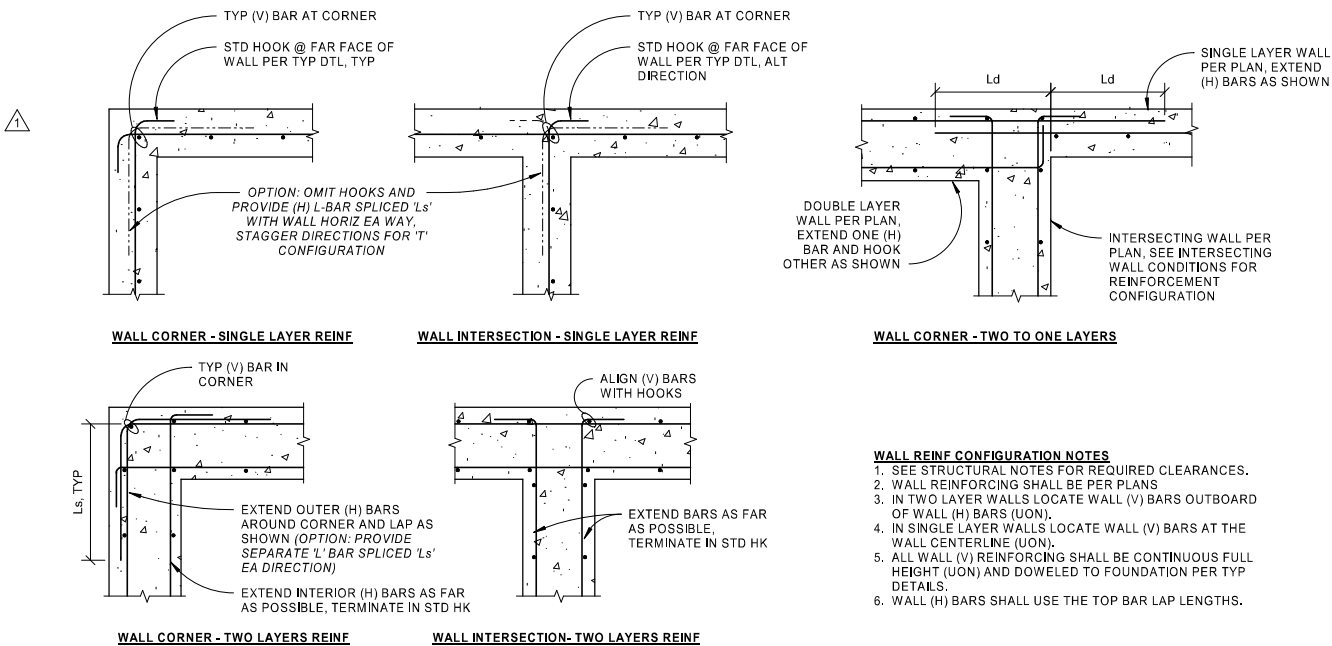
8 TYP TRENCH IN (E) SLAB FOR SUB-FLOOR PIPE
1 1/2" = 1'-0"



- NOTES**
1. FOR CONC STRENGTHS NOT INDICATED, USE NEXT LOWEST TABULATED f_c VALUE.
 2. PROVIDE 2 1/2" SIDE COVER AND 2" END COVER FOR HOOKS.
 3. TABULATED VALUES ARE APPLICABLE FOR GRADE 60 UNCOATED REINF BARS.

BAR SIZE	STANDARD HOOK EMBEDMENT (Ldh)	
	CONC DESIGN STRENGTH (f_c)	
#3	4,500 PSI	
#4	6"	
#5	7"	
#6	9"	
#7	10"	
#8	12"	
#8	14"	

5 TYP CONCRETE REINFORCING STANDARD HOOKS
3" = 1'-0"



2 TYP CONCRETE WALL REINFORCING AT CORNERS & INTERSECTIONS
3/4" = 1'-0"

- WALL REINF CONFIGURATION NOTES**
1. SEE STRUCTURAL NOTES FOR REQUIRED CLEARANCES.
 2. WALL REINFORCING SHALL BE PER PLANS
 3. IN TWO LAYER WALLS LOCATE WALL (V) BARS OUTBOARD OF WALL (H) BARS (UON).
 4. IN SINGLE LAYER WALLS LOCATE WALL (V) BARS AT THE WALL CENTERLINE (UON).
 5. ALL WALL (V) REINFORCING SHALL BE CONTINUOUS FULL HEIGHT (UON) AND DOWELED TO FOUNDATION PER TYP DETAILS.
 6. WALL (H) BARS SHALL USE THE TOP BAR LAP LENGTHS.



 [Seattle, WA 98108, USA](#)

EQUIPMENT AND TOOLS / GENERATORS AND ACCESSORIES / GENERATOR - 499KW AND SMALLER

CAT CLASS: 0090105

120 kW Diesel Generator



\$735

1 DAY

\$1,975

1 WEEK

\$4,620

4 WEEK

Rates for **Seattle, WA 98108, USA**

Subject to availability. The displayed rates may vary by location and do not include the taxes or optional charges that may be selected later in the checkout process.

[Request quote](#)



Overview

A 120 kW diesel generator provides ideal power solutions for temporary, standby and emergency power requirements. All models feature quiet operation (60 dBA or lower at full load) and onboard fuel capacity of 24-hour minimum run times. Multiple voltages are easily configured, simultaneous single and 3-phase power output. During a declared state of emergency and/or a pending or existing natural disaster, all rentals of this generator type will be billed at a one-week minimum rental period for 24-hours of usage per day. The one-week minimum rental at 24-hour usage rates does not apply to pre-existing generator rentals.

Specifications

Products are subject to availability.

Make	MQ Power	
Model	120 kW Diesel Generator	
Consump# Full Load	9 GPH	
Dual Fuel_(NG/LP)	false	
Engine	John Deere 180 HP	
Hertz	60	
Onboard Fuel Cap	269 gal.	
Operating Weight	9784 lbs	
Overall Dimensions	202"L x 83"W x 89"H	
Prime Rating	120 kW	
Run Time Full Load	30 hrs.	

Safety information

Safety Requirements



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